

Tugas 2: Laporan Praktikum Mandiri 2

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The screenshot shows a Jupyter Notebook interface with two cells. Cell [6] contains a pandas DataFrame manipulation code snippet that filters a DataFrame 'df' to include only specific columns: 'season', 'yr', 'mnth', 'holiday', 'weekday', 'workingday', 'weathersit', 'temp', 'atemp', 'hum', 'windspeed', and 'cnt'. The result is stored in 'df1' and its first five rows are displayed as a table. Cell [8] contains a scikit-learn code snippet that imports 'train_test_split', defines the target variable 'y' as 'cnt' and the feature matrix 'X' with the selected columns, and then splits the data into training and testing sets using an 80:20 ratio. The lengths of the resulting training and testing sets are printed.

```
[6]
✓ 0s
df1 = df[['season', 'yr', 'mnth', 'holiday', 'weekday',
          'workingday', 'weathersit', 'temp', 'atemp', 'hum', 'windspeed', 'cnt']].copy()
df1.head()
```

	season	yr	mnth	holiday	weekday	workingday	weathersit	temp	atemp	hum	windspeed	cnt
0	1	0	1	0	6	0	2	0.344167	0.363625	0.805833	0.160446	985
1	1	0	1	0	0	0	2	0.363478	0.353739	0.696087	0.248539	801
2	1	0	1	0	1	1	1	0.196364	0.189405	0.437273	0.248309	1349
3	1	0	1	0	2	1	1	0.200000	0.212122	0.590435	0.160296	1562
4	1	0	1	0	3	1	1	0.226957	0.229270	0.436957	0.186900	1600

Next steps: [Generate code with df1](#) [New interactive sheet](#)

```
[8]
✓ 0s
from sklearn.model_selection import train_test_split

# Tentukan variabel dependen dan independen
y = df1['cnt']
X = df1[['season', 'yr', 'mnth', 'holiday', 'weekday',
          'workingday', 'weathersit', 'temp', 'atemp', 'hum', 'windspeed']]

# Bagi data menjadi training dan testing (80:20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Jumlah data training :", len(X_train))
print("Jumlah data testing  :", len(X_test))
```

Jumlah data training : 584
Jumlah data testing : 147

- Cell 6: Menyimpan hanya kolom yang relevan untuk model prediksi (cnt dan faktor yang memengaruhinya).
- Cell 8: Memisahkan data menjadi:
 - X** = variabel independen (faktor penyebab)
 - y** = variabel dependen (cnt)
 - 80% data untuk pelatihan, 20% untuk pengujian.

[9]
✓ 0s

```
from sklearn.linear_model import LinearRegression

# Buat dan latih model regresi linear
model = LinearRegression()
model.fit(X_train, y_train)
```

LinearRegression ⓘ ?

LinearRegression()

[10]
✓ 0s

```
import numpy as np
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

# Prediksi data uji
y_pred = model.predict(X_test)

# Evaluasi model
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print("Koefisien tiap variabel:")
for nama, coef in zip(X.columns, model.coef_):
    print(f" {nama:12s} : {coef:.3f}")

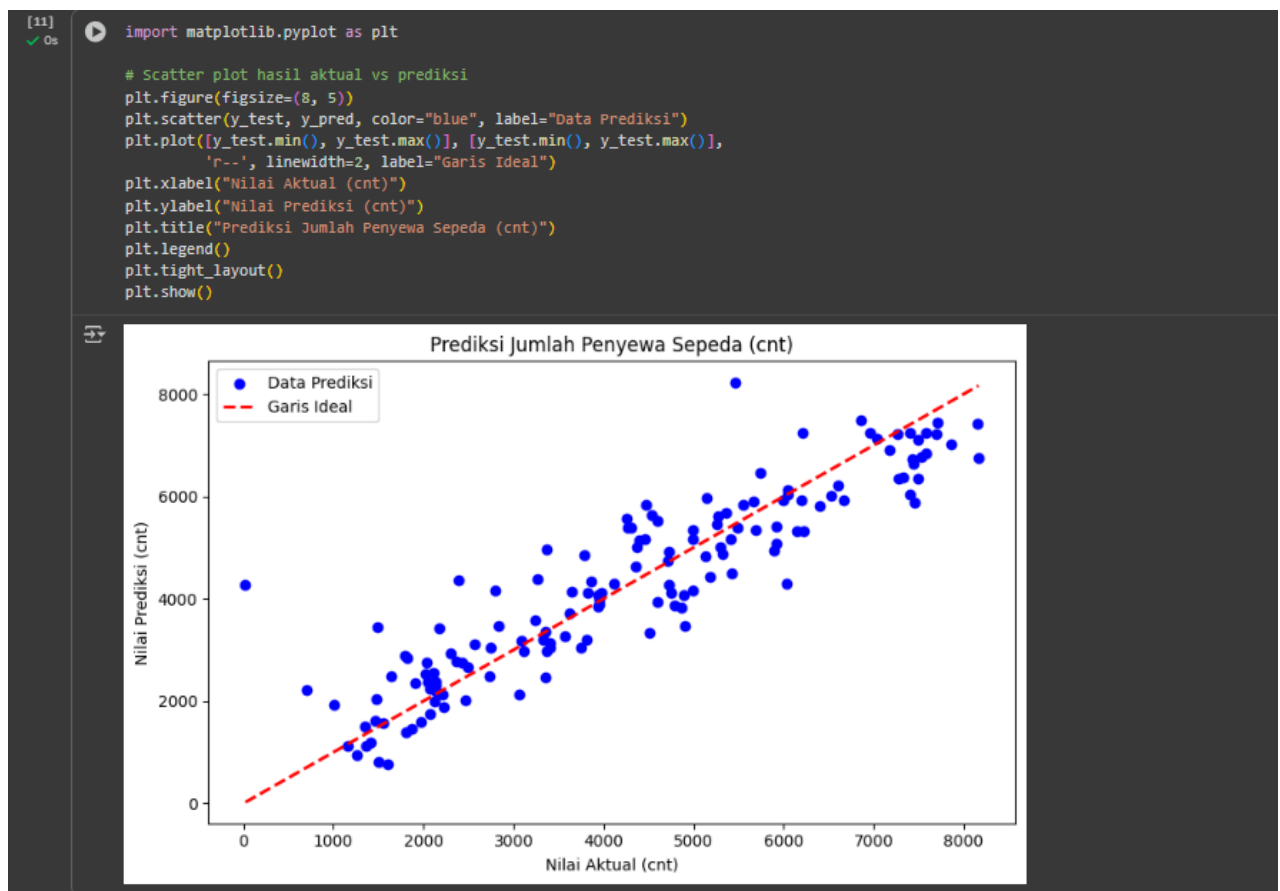
print("\nIntersep:", model.intercept_)
print(f"R² (test): {r2:.3f}")
print(f"MAE: {mae:.2f}")
print(f"MSE: {mse:.2f}")
print(f"RMSE: {rmse:.2f}")
```

Koefisien tiap variabel:

season	: 524.723
yr	: 2023.998
mnth	: -38.445
holiday	: -391.551
weekday	: 72.937
workingday	: 160.805
weathersit	: -632.856
temp	: 2097.248
atemp	: 3488.042
hum	: -865.439
windspeed	: -2080.540

Intersep: 1248.3209284778172
R² (test): 0.828
MAE: 617.39
MSE: 691035.01
RMSE: 831.29

- Cell 9: Membuat model regresi linear dan melatihnya dengan data training.
- Cell 10: Melakukan prediksi pada data uji dan menghitung metrik performa:
R² → seberapa baik model menjelaskan data
MAE, MSE, RMSE → tingkat kesalahan prediksi



- Cell 11: Menampilkan grafik **aktual vs prediksi**; garis merah menunjukkan hasil ideal (prediksi = aktual).

```
hasil = pd.DataFrame({
    "Cnt Aktual": y_test.to_numpy(),
    "Cnt Prediksi": y_pred
})

# Hitung selisih dan akurasi
hasil["Selisih (cnt)"] = hasil["Cnt Prediksi"] - hasil["Cnt Aktual"]
denom = hasil["Cnt Aktual"].replace(0, np.nan)
hasil["Akurasi (%)"] = (1 - (hasil["Selisih (cnt)"].abs() / denom)).clip(lower=0, upper=1) * 100

hasil.head(10)
```

	Cnt Aktual	Cnt Prediksi	Selisih (cnt)	Akurasi (%)
0	6606	6220.416612	-385.583388	94.163134
1	1550	1571.605763	21.605763	98.606080
2	3747	3043.967028	-703.032972	81.237444
3	6041	4307.202484	-1733.797516	71.299495
4	7538	6765.659072	-772.340928	89.754034
5	7264	7231.536106	-32.463894	99.553085
6	1605	774.904147	-830.095853	48.280632
7	2209	2139.606601	-69.393399	96.858606
8	7499	7112.745605	-386.254395	94.849255
9	5743	6458.992169	715.992169	87.532785

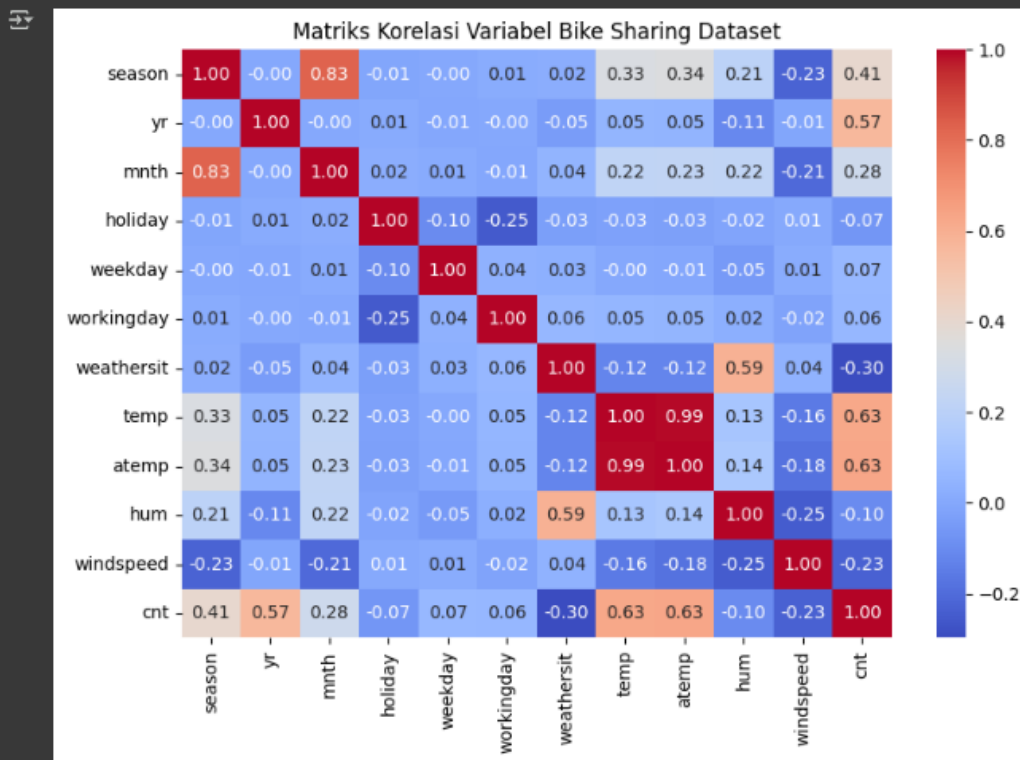
Next steps: [Generate code with hasil](#) [New interactive sheet](#)

- Cell 12: Membandingkan nilai aktual dan prediksi. Menambahkan kolom **selisih** dan **akurasi per baris** dalam persen.

[13]
✓ 1s

```
import seaborn as sns

plt.figure(figsize=(9, 6))
sns.heatmap(df1.corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Matriks Korelasi Variabel Bike Sharing Dataset")
plt.show()
```



[15]
✓ 0s

```
hasil.to_csv(path + '/data/hasil_prediksi_bikesharing.csv', index=False)
print("✓ File hasil_prediksi_bikesharing.csv telah disimpan di Google Drive kamu!")
```

✓ File hasil_prediksi_bikesharing.csv telah disimpan di Google Drive kamu!

- Cell 13: Menampilkan hubungan antar variabel dengan **heatmap korelasi**. Semakin mendekati 1, semakin kuat hubungan antar variabel.
- Cell 15: Menyimpan tabel hasil prediksi ke file .csv di Google Drive.

Link GitHub: <https://github.com/CeperMail/MachineLearning.git>